

Recap

① Cumulative distribution Function

$$F_X(x) = P(X \leq x)$$

Properties

① Monotonicity

If $x < y$

$$F_X(x) \leq F_X(y)$$

$$\textcircled{2} \lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\lim_{x \rightarrow \infty} F_X(x) = 1$$

Examples

① Discrete RV

$$X_0 = \{x_1, x_2, \dots\}$$

$$P_X(x_i) = F_X(x_i) - F_X(x_{i-1})$$

$$x_1 < x_2 < \dots$$

② $\Omega = [0, 1]$

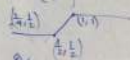
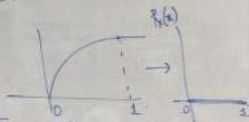
$$X(\omega) = \omega^2$$

$$F_X(x) = \sqrt{x} \quad x \in [0, 1]$$

②a

$$P_X(x) = F_X(x)$$

$$-F_X(x) \quad 0 \quad \frac{1}{4}$$



Right Continuity

$$\textcircled{3} \quad F_X(x^+) = \lim_{\epsilon \rightarrow 0^+} F_X(x+\epsilon) \\ = \lim_{n \rightarrow \infty} F_X\left(x + \frac{1}{n}\right)$$

$$x \quad \cdot \quad \cdot \quad \cdot \quad x + \frac{1}{2} \quad x + 1$$

←

$$F_X\left(x + \frac{1}{n}\right) = P\left(X \leq x + \frac{1}{n}\right) \\ = P\left(\left\{\omega : X(\omega) \leq x + \frac{1}{n}\right\}\right)$$

$$A_n = X^{-1}\left(\left(-\infty, x + \frac{1}{n}\right]\right) \\ = \left\{\omega : X(\omega) \leq x + \frac{1}{n}\right\}$$

$$F_X(x^+) = \lim_{n \rightarrow \infty} P(A_n)$$

$$A_1 \supseteq A_2 \supseteq A_3 \supseteq \dots$$

$$= P\left(\bigcap_{n=1}^{\infty} A_n\right) \quad \left[\text{Using Continuity Corollary 2} \right]$$

$$= P\left(X^+(-\infty, x]\right)$$

$$= F_X(x)$$

$$F_X(x^-) = \lim_{n \rightarrow \infty} F_X(x - \frac{1}{n})$$

Show that

(Exercise) $F_X(x^-) = P(X < x)$

Expec

$$P\left(\bigcup_{n=1}^{\infty} A_n\right)$$

Corollary 2

$$A_1 \supseteq A_2 \supseteq \dots$$

$$P\left(\bigcap_{n=1}^{\infty} A_n\right)$$

Expectation (Mean)

Suppose you run an experiment N times and you see outcome x , n_x # of times.

Average = $\frac{\sum_{x \in \mathcal{X}_0} x n_x}{N}$

relative frequency

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} P\left(\bigcup_{i=1}^n A_i\right)$$

Corollary 2:

$$A_1 \supseteq A_2 \supseteq \dots$$

$$P\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} P(A_n)$$

$$E[X] = \sum_{x \in \mathcal{X}_0} x P_X(x)$$

$E[X]$ is well defined if it absolutely converges.

Absolutely converges

$$\Leftrightarrow \sum_{x \in X_0} |x| P_X(x) < \infty$$

Examples:

① $|X_0| < \infty$, X_0 is finite

$$\hat{x}_{\max} = \max_{x \in X_0} |x|$$

$$\begin{aligned} \sum_{x \in X_0} |x| P_X(x) &\leq \sum_{x \in X_0} \hat{x}_{\max} P_X(x) \\ &= \hat{x}_{\max} < \infty. \end{aligned}$$

$$x_{\min} \leq E[X] \leq x_{\max}$$

$$\Rightarrow \sum_{x \in \mathcal{X}_0} x_{\min} P_X(x) \leq \sum_{x \in \mathcal{X}_0} P_X(x) x \leq \sum_{x \in \mathcal{X}_0} x_{\max} P_X(x)$$

$$\Rightarrow x_{\min} \leq E[X] \leq x_{\max}$$

$$(2) \quad \mathcal{X}_0 = \{2^n \mid n=1, 2, \dots\}$$

$$P(X=2^n) = \frac{1}{2^n} \quad n=1, 2, \dots$$

$$E[X] = \sum_{n=1}^{\infty} 2^n P_X(2^n) = \sum_{n=1}^{\infty} 1 \quad (\text{doesn't converge})$$

$$\textcircled{3} X_0 = \{ (-2)^n \mid n=1, 2, \dots \}$$

$$P(X = (-2)^n) = \frac{1}{2^n} \quad n=1, 2, \dots$$

$$\sum_{n=1}^{\infty} |(-2)^n| \frac{1}{2^n} = \sum_{n=1}^{\infty} 2^n \times \frac{1}{2^n}$$

Doesn't converge absolutely

$$\sum_{n=1}^{\infty} P(X = (-2)^n) \frac{1}{2^n}$$

$$= \sum_{n=1}^{\infty} (-2)^n \frac{1}{2^n} = \sum_{n=1}^{\infty} (-1)^n$$

doesn't
converge)

①

$$S_n = \sum_{i=1}^n a_i$$

$$s_1 \quad s_2 \quad s_3 \quad \dots$$

To show $\sum_{i=1}^{\infty} a_i$ is well defined

$$-1 \quad 0 \quad -1 \quad 0 \quad \dots \quad (\text{doesn't converge})$$

Properties of expectation

① Expectation of a function of a RV

Let X be a RV with PMF $P_X(x)$.

and $Y = g(X)$

$$\text{then } E(Y) = \sum_{x \in \mathcal{X}} g(x) P_X(x)$$

$$E[Y] = \sum_{x \in \mathcal{X}_0} g(x) P_X(x)$$

$$E[Y] = \sum_{y \in \mathcal{Y}} P_Y(y) y$$

$$= \sum_{y \in \mathcal{Y}} y \sum_{x: g(x)=y} P_X(x)$$

$$= \sum_{y \in \mathcal{Y}} \sum_{x: g(x)=y} y P_X(x)$$

$$= \sum_{y \in \mathcal{Y}, x: g(x)=y} g(x) P_X(x)$$

$$= \sum_{x \in \mathcal{X}_0} g(x) P_X(x)$$



$$\textcircled{2} \quad E[aX+b] = aE[X] + b$$

$$\begin{aligned} E[aX+b] &= \sum_{x \in \mathcal{X}_0} (ax+b) P_X(x) \\ &= a \sum_{x \in \mathcal{X}_0} x P_X(x) + b \underbrace{\sum_{x \in \mathcal{X}_0} P_X(x)}_1 \\ &= aE[X] + b \end{aligned}$$

$\textcircled{3}$ If X is a R.V takes non-negative values
and if $E[X]=0 \Rightarrow X=0$ w.p 1 $\iff X>0$ w.p 0

Otherwise i.e., if $X>0$ w.p >0

$$\begin{aligned} E[X] &= \sum_{x \in \mathcal{X}_0} x P_X(x) = \sum_{x \neq 0} x P_X(x) \\ &> 0 \end{aligned}$$

Moments of a RV

$$E[X^n] = \sum_{x \in \mathcal{X}} x^n P_X(x)$$

$$\text{Let } g(x) = x^n$$

Variance

$$\text{Var}(X) = E[(X - E[X])^2] \geq 0$$

① $\text{Var}(X) = 0 \iff X = E[X] \text{ w.p. } 1$

$Y = (X - E[X])^2$ is a non-negative R.V.

$$E[Y] = 0 \implies Y = 0 \text{ w.p. } 1$$

ie, $(X - E[X])^2 = 0 \text{ w.p. } 1$

$$\Rightarrow X = E[X] \text{ w.p. } 1$$

$$\textcircled{2} \text{ Var}(aX+b) = a^2 \text{Var}(X)$$

$$\begin{aligned} \text{Var}(aX+b) &= E[(aX+b - E[aX+b])^2] \\ &= E[(aX+b - aE[X] - b)^2] \quad (\text{property 2 of expected value}) \\ &= E[a^2 (X - E[X])^2] \\ &= a^2 E[(X - E[X])^2] \\ &= a^2 \text{Var}(X) \end{aligned}$$

$$\textcircled{3} \text{ Var}(X) = E[X^2] - (E[X])^2$$